



National University of Computer and Emerging Sciences

Laboratory Manuals

for

Computer Networks - Lab

(CL -3001)

Department of Computer Science

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Lab Manual 04

Objective:

- Analyzing FTP and ICMP packets using Wireshark.
- Capturing and analyzing the structure of the User Datagram Protocol (UDP).

Lab Statement 1: Capturing FTP packets using Wireshark

Step 1: Start a Wireshark capture

Close all unnecessary network traffic to limit the amount of traffic during the Wireshark capture. Start the Wireshark capture.

Step 2: Connect and Download using FileZilla

For this lab, we will use FileZilla, a modern FTP client. The standard Windows command-line FTP client is very old and does not support passive (PASV) mode, which can cause connection issues with modern servers. FileZilla fully supports PASV mode and is easier to use.

Open FileZilla and use the following credentials to connect to the server:

- **Host:** 4.224.76.5
- **Username:** bcs5c (Select according to your Section i.e. bcs5a, bcs5g)
- **Password:** Fast@123456789

Once connected, perform some actions like listing directories, creating a folder, and downloading a file (e.g., `Test File.txt`).

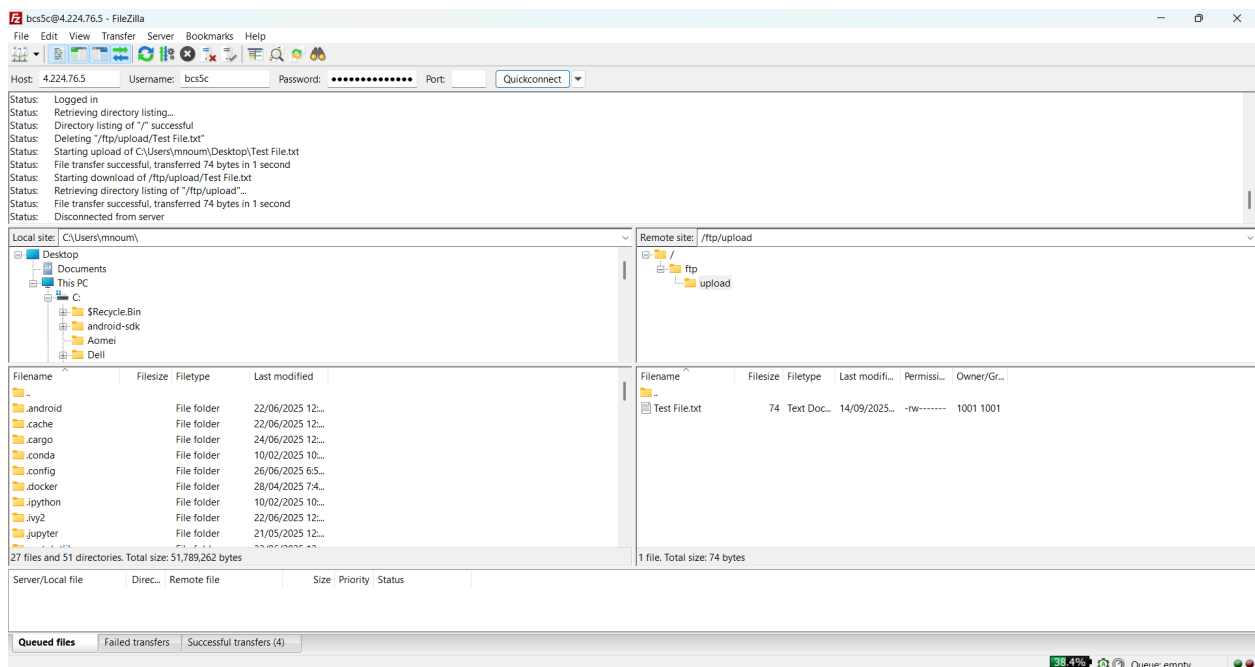


Figure 1: FileZilla interface showing a successful FTP connection and file transfer.

Step 3: Stop the Wireshark capture

Stop the packet capture.

Step 4: View the Wireshark Main Window

To limit the data for analysis, type `tcp and ip.addr == 4.224.76.5` in the Filter bar. This is the IP address of the FTP server you connected to.

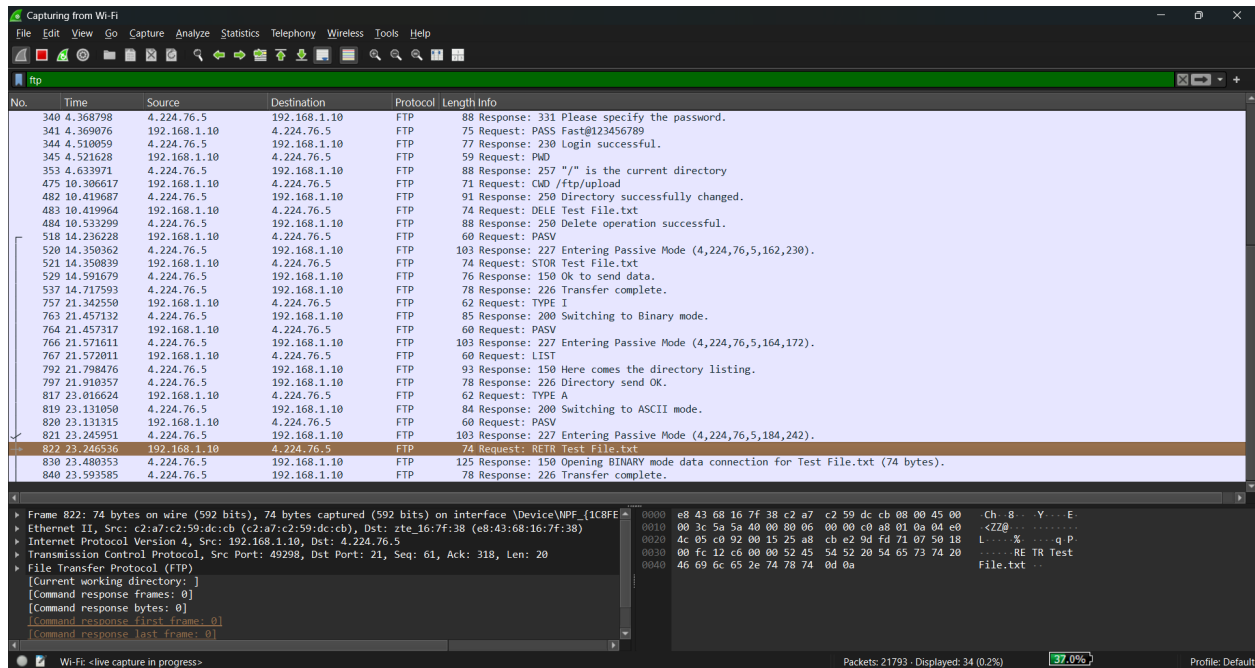


Figure 2: Wireshark capture showing the FTP session after downloading a file.

Step 5: Analyze Your Capture

Carefully analyze the packets from your live FTP session in Wireshark to answer the following questions:

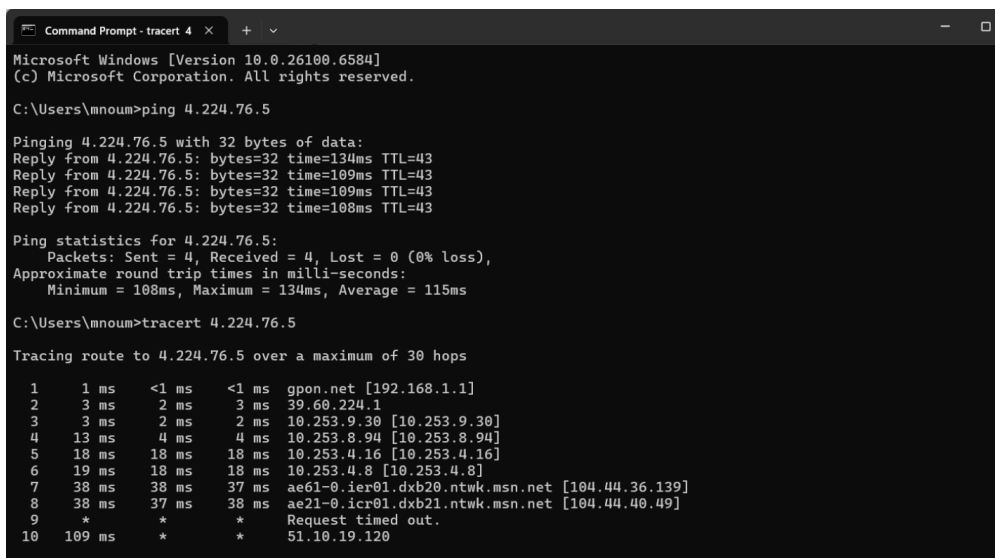
1. What is the TCP port number for the FTP control connection? Describe the initial messages exchanged between your client and the server to establish this connection (e.g., TCP handshake, FTP welcome message).
2. Find the packets where your client sends the username and password. Are these credentials encrypted? What FTP commands are used for this (e.g., USER, PASS)?
3. FTP uses a separate data connection to transfer files and list directories. Find the packets that establish this second connection. Is the TCP port for this data connection the same as the control connection? Explain your observation.
4. Select the packets corresponding to your file download. What protocol is indicated in the "Protocol" column for these packets (FTP or FTP-DATA)? What does this tell you about how FTP separates commands from file contents?

5. Look at the FTP commands sent by your client (e.g., listing a directory). What command does the client send, and what is the server's response code and message?
6. Find the packet where your client ends the session. What FTP command is used to terminate the connection (e.g., QUIT)? What is the server's final response code and message?

Lab Statement 2: Analyzing ICMP Packets using Wireshark

Steps:

1. Start a new Wireshark capture.
2. Open a command prompt or terminal.
3. Run the command `ping 4.224.76.5` to send ICMP echo requests to the server.



```
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mnoum>ping 4.224.76.5

Pinging 4.224.76.5 with 32 bytes of data:
Reply from 4.224.76.5: bytes=32 time=134ms TTL=43
Reply from 4.224.76.5: bytes=32 time=109ms TTL=43
Reply from 4.224.76.5: bytes=32 time=109ms TTL=43
Reply from 4.224.76.5: bytes=32 time=108ms TTL=43

Ping statistics for 4.224.76.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 108ms, Maximum = 134ms, Average = 115ms

C:\Users\mnoum>tracert 4.224.76.5

Tracing route to 4.224.76.5 over a maximum of 30 hops
  0  1 ms  <1 ms  <1 ms  gpon.net [192.168.1.1]
  1  3 ms  2 ms  3 ms  39.68.224.1
  2  3 ms  2 ms  2 ms  10.253.9.30 [10.253.9.30]
  3  13 ms  4 ms  4 ms  10.253.8.94 [10.253.8.94]
  4  18 ms  18 ms  18 ms  10.253.4.16 [10.253.4.16]
  5  19 ms  18 ms  18 ms  10.253.4.8 [10.253.4.8]
  6  38 ms  38 ms  37 ms  ae61-0.ier01.dxb20.ntwk.msn.net [104.44.36.139]
  7  38 ms  37 ms  38 ms  ae21-0.icr01.dxb21.ntwk.msn.net [104.44.40.49]
  8  * * * Request timed out.
  9  109 ms * * 51.10.19.120
```

Figure 3: Example output of the ping command.

4. Next, run the command `tracert 4.224.76.5` to trace the route to the server. This also uses ICMP messages.
5. Stop the Wireshark capture and filter for `icmp` packets.
6. Analyze the packets to answer the questions below.

Analysis Questions:

1. Are ICMP messages sent over UDP or TCP?
2. What is the link-layer (Ethernet) address of the host?
3. What kind of request is sent through these ICMP packets?
4. How many requests are sent by the host?
5. What are the IP addresses of your host and the destination host?

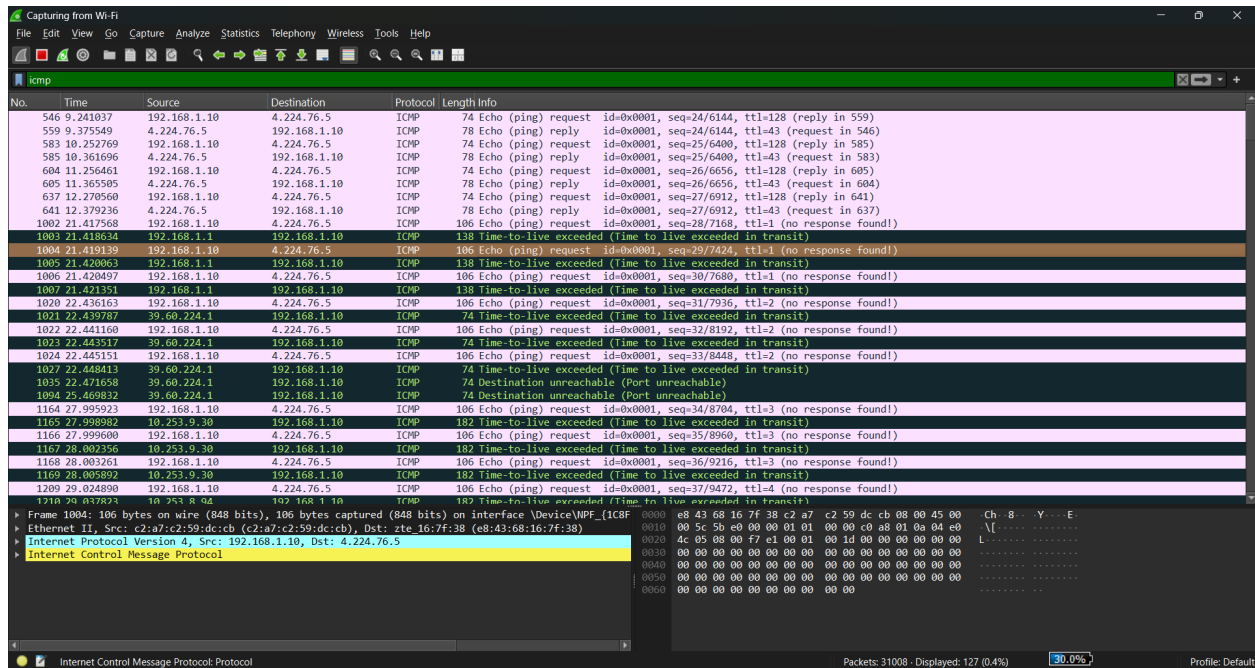


Figure 4: Wireshark window displaying captured ICMP packets.

6. Why doesn't an ICMP packet have source and destination port numbers?
7. What values differentiate an ICMP request message from a reply message?
8. Examine a ping request packet. What are the ICMP type and code numbers? How many bytes are the checksum, sequence number, and identifier fields?
9. Examine the corresponding ping reply packet. What are its ICMP type and code numbers?
10. Examine packet no. 56. What are its ICMP type and code numbers?
11. Why is an IP and TCP Header included in some ICMP messages? What do these headers depict?

Lab Statement 3: Analyzing UDP Packets using Wireshark

The User Datagram Protocol (UDP) is a simple, connectionless transport protocol. We will use the `nslookup` tool to generate UDP (DNS) traffic for analysis.

Steps:

1. Start capturing packets in Wireshark.
2. Open a command prompt and run `nslookup` for a hostname like `www.nyu.edu`.
3. Stop the capture and filter for `udp` packets.

Analysis Questions:

Examine the first UDP packet (DNS query) and its corresponding reply to answer the following:

```
[newworld.cs.umass.edu> nslookup www.nyu.edu
Server:          128.119.240.1
Address:         128.119.240.1#53

Non-authoritative answer:
www.nyu.edu      canonical name = WEB.GSLB.nyu.edu.
Name:   WEB.GSLB.nyu.edu
Address: 216.165.47.12
Name:   WEB.GSLB.nyu.edu
Address: 2607:f600:1002:6113::100
```

Figure 5: Example of running the nslookup command.

1. How many fields are in the UDP header, and what are their names and lengths in bytes?
2. The value in the UDP "Length" field refers to the length of what?
3. What is the maximum possible size of a UDP payload? What is the largest possible source port number?
4. What is the protocol number for UDP (found in the IP header)?
5. Describe the relationship between the source and destination port numbers in the query and reply packets.